



**DATA CENTRES: POLICY**  
**A BALANCING ACT FOR ECONOMY & SOCIETY**  
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# COMMITMENT DAY

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# Data Centres – Expert panel - Policy

This session recognises major changes in energy and environmental policy (especially concerning data centres), and builds on the current situation about Data Centres.

Relevant bodies include **Ministry of Energy (NEPC/EPPO, ERC (Min Energy) BOI, NBTC, Min Industry, Data Centre Policy Committee.**

It considers international and regional trends, the value of data centres, the wider purpose in serving the Digital Economy, AI demand and other needs, how efficiencies may be reflected, regional sharing of resources eg ASEAN Power Grid.

What should the policies be and what considerations should they reflect.

To assist with base information about high level status of current policies, an opening backgrounder is provided

# Data Centres – Expert panel - Policy

**MK Balaji** - Strategic Advisory Board - Energy Regulators Regional Association (ERRA)

**Dr Bordin Vongvitayapirom** , Senior Manager, Energy, Resources and Industrial SME, Deloitte Thailand

**Dr Chanchai Amornvipas** Director of Energy Regulatory Development and Strategic Execution Department. Office of Energy Regulatory Commission (ERC)

**Asst Professor Nondh Nuchmorn**; Faculty of Economics Thammasat University

**Moderator: Mr Bob Fox** – Chair Digital Economy/ICT JFCCT and JFCCT Policy Co-Ordinator



## **DATA CENTRES: POLICY**

### **A BALANCING ACT FOR ECONOMY & SOCIETY**



**M.K Balaji**

Member of the Advisory Board  
Energy Regulators Regional  
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**Moderator Bob Fox**

Chair, Digital Economy  
and Policy Co-Ordinator  
Joint Foreign Chambers  
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# Question themes – Policy panel

**THEME ONE: CURRENT POLICIES OR IN PIPLINE**

**THEME TWO: What should the future landscape be?  
What policies are essential?**



# Data Centres – fast developing regulation

(this slide + next five: opening background)

A global phenomenon - energy, water, noise, in some cases land use, low employment.

Now several agencies shape policy and regulation:

**NEPC/EPPO, ERC (Min Energy) BOI, NBTC, Min Industry, Data Centre Policy Committee (DCPC).**  
**DCPC expected to clarify all soon**

Following slides consider BOI, NEPC, NBTC; see Dr Chanchi (ERC)

2024 – Direct PPA proposal of 2,000 MW

May 2025 – BOI policy outline

Feb 2026 – BOI ‘rules’ with a change in March.





## Investment Promotion for Data Center

### Criteria for Investment Promotion

#### Services

- Must provide complementary services (e.g. Server co-location, Managed service, Customer's server backup service, Disaster recovery service (DRS))

#### Capacity

- IT load capacity must not be less than **2 MW**

#### Connectivity

- Must have a main hi-speed telecommunications system linking the data center with domestic & international telecommunication centers at least 4 systems (The domestic telecommunication system must have a speed  $\geq 10$  Gbps each, for  $\geq 3$  systems, and the total speed of the whole system  $\geq 60$  Gbps)

#### Cooling System & Fire Protection

- Must have a highly efficient air conditioning system with backup system
- Must have fire prevention system for the entire area

#### Others

- Must submit an efficient and appropriate water use management plan for data center
- Must employ Thai nationals in senior management and/or specialist positions, accounting for at least 50% of the total personnel in such positions within 3 years.
- Must submit an action plan for creating benefits for the country, such as organizing training and developing joint curricula with educational institutions

#### Redundancy

- Must be **Concurrently Maintainable**
- Must have backup independent distribution paths in electricity distribution system
- Must have a generator which is capable of supporting the entire electricity needs of the data center with a backup generator that can support the need for electricity when one of the generators does not function properly
- Must have equipment or backup system for UPS IT cooling and UPS cooling
- Must have a failure prevention system to prevent risks from damage or any malfunction in any equipment

#### Security

- Must have a 24-hour security system

#### Certification

- Must be certified with **ISO/IEC 27001 (Data Center)** before exercising the incentive of corporate income tax exemption and before the full operation deadline
- **Must obtain a certification letter from the Energy Regulatory Commission (ERC) prior to submitting the BOI application.**

### Incentive

A2-A4



#### CIT Exemption

**8 or 5 (in EEC) years**  
(if  $PUE \leq 1.3$ )

**5 or 3 (in EEC) years**  
(if  $PUE > 1.3$ )



#### Exemption of Import duty on machinery



#### Non-tax incentives



About June : Finance Minister Ekniki asked BOI to revise plans.

Late July/Early August – high level changes:

Electricity supply and energy readiness, including the transition towards greater use of clean energy.

Water-resource management - projects must operate without competing with surrounding communities for water.

Environmental protection and pollution controls, including noise.

Tangible benefits for Thailand; efficient use of national resources but supporting digital and advanced-technology investment.

# NEPC – 15 July – some key points

Household tariff at THB 3 / unit (1 kilowatt-hour (kWh))

Direct PPA expanded to clean-energy industries (not just Data Centres)

Special tariff for Data Centres eg 5-6 THB + wheeling and other charges

Intent: prevent these additional costs from being passed on to general electricity users.

Large DCs – bond

Water management plan

Community based solar - ground max 10MW; max 1500 MW and limits on number per operator

***BUT – still issues about decentralisation and no real market based energy price***

**26 have NBTC cat 1 licence under Telecoms Business Act**

**5 rules recently released by NBTC**

- 1) Location depends on infra (zoning)
- 2) Electricity and water supply contracts in place
- 3) Dedicated systems for telecoms, Electricity, water (means)
- 4) e Waste disposal plan, clean energy and % clean energy
- 5) Min 50% infra must be sourced from Thai businesses to distribute through the supply chain

But NBTC's proposed change to cat 3 licence (no majority foreign ownership) does not appear.





# Data Centres end-to-end issues – JFCCT Digital Economy/ICT consistently since Dec 2024

## DATA

Data Governance / collection

Analysis / Processing

## PROCESSING

Photonics, chips

LLMs vs SLMs

AI model consumption

## RESOURCES

Energy / Renewable Energy PUE

Special RE tariffs

Floating solar

Water usage WUE

Possible national rationing / priority

## CONSTRUCTION

Wood

Under water

## NATIONAL POLICY

Over promotes DCs ?

NBTC – cat 3 licence?

BOI conditions

Data Localisation

Location and proximities/ zoning

NEPC

DC Policy Committee

## REGIONAL

ASEAN Power Grid

ASEAN co-operation

Other regional practice – conditions

APDCA – Sustainable Digital Infrastructure Accord

[\(SDIA\);](#)

## MODE

Terrestrial

Orbital

Ocean

Digital Economy policy – the regional hub will not be decided by the economy with most<sup>12</sup> Data Centre space.



## DATA CENTRES: POLICY

### A BALANCING ACT FOR ECONOMY & SOCIETY



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# Data Centres – Expert panel - Policy

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Following three slides –

MK Balaji -Strategic Advisory Board - Energy Regulators Regional Association (ERRA)



# Policy has advanced — success now goes beyond megawatts

Thailand’s promoted pipeline exceeds the original pilot; leading regimes link scarce capacity to location, additionality, flexibility and resource efficiency.

42

promoted projects  
2024–2026

3.4 GW

combined IT load  
before PUE overhead

≈750bn

THB investment  
promoted pipeline

## IT IS CONTINUOUSLY DEVELOPING

15 JUL

**Policy scope broadened** DPPA beyond data centres; separate DC tariff, bond and fund direction.

7–21 AUG

**Grid plan made visible** THB 31.05bn TSLC programme under ERC consultation.

14 AUG

**National committee in force** Energy, digital, investment, water and planning in one loop.

POLICY APPROVED • COMMERCIAL OPERABILITY STILL TO PROVE

## WHAT INTERNATIONAL PRACTICE ADDS - *import the design principles — not another country’s thresholds*

IRELAND

**Location + adequacy + additionality**

Connection reflects constraints; matching generation/storage; additional renewables with a glide path.

EUROPEAN  
UNION

**Measure energy and water performance**

Mandatory reporting supports a common database and an emerging rating framework. PUE and WUE become policy evidence, not marketing claims.

SINGAPORE / EU

**Capacity linked to resource performance**

PUE/WUE, green energy and transparent reporting make energy and water part of allocation.

IEA / GLOBAL

**Trade flexibility for faster access**

Queue reform, non-firm connections, demand response and grid-interactive BESS.

## THAILAND SUCCESS TEST

01

**Available + reliable**

Capacity where and when needed

02

**Additional + clean**

Traceable renewables; credible glide path

03

**Fair + auditable**

Cost causation; no double recovery

04

**Resource-efficient**

PUE/WUE and community impacts

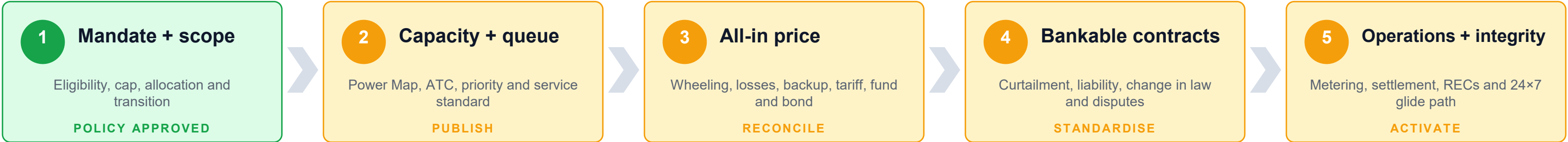
*Renewables requirement: a credible forward plan from day one — annual matching first, tightening toward 24×7 as the market matures.*





# Open in phases — make every phase complete enough to transact

A bankable DPPA is a connected chain of rules, prices, contracts and operations; one missing gate can stop the deal.



NO-DOUBLE-RECOVERY TEST

## Cost-reflective ≠ punitive

Generator energy + wheeling + losses/imbalance + backup/ancillary + special tariff/fund + bond

Separate baseline from incremental cost; credit customer-funded assets and verified flexibility; stage and refund the bond; publish one auditable calculator and one cost-of-service bridge.

FUTURE LANDSCAPE

## A decentralised, regulated market — not a permanent carve-out

- **Transparent access** Neutral queue, capacity map and connection options
- **Flexible participation** BESS, demand response and non-firm service earn value
- **Explicit public policy** Consumer protection and social contributions stay visible

12-MONTH DELIVERABLES

Roadmap • Power Map + queue • all-in calculator • standard agreements • flexibility + integrity framework

Winning proposition: predictable access to reliable, additional clean power at an auditable price — without shifting incremental cost to ordinary consumers.



# Quantified stack ≈ THB 4.78/kWh — plus material TBD items

Assumptions: 95% load factor • 69 kV+ delivery • 1.664 TWh/year • solar+BESS gate-price proxy • excludes VAT and site-specific reinforcement.

| COMPONENT                   |                                                            | THB/kWh | THB bn/yr | BASIS   |
|-----------------------------|------------------------------------------------------------|---------|-----------|---------|
| Solar + BESS gate price     | Thai FiT reference used as proxy; not 24×7 firm supply     | 2.8331  | 4.715     | PROXY   |
| Wheeling — 69 kV+           | Transmission 0.6868 + distribution 0.4350                  | 1.1218  | 1.867     | DRAFT   |
| System security / ancillary | Reserve + voltage + frequency support                      | 0.4978  | 0.829     | DRAFT   |
| Policy expense              | Sep–Dec 2025 reference; designed to reset                  | 0.1447  | 0.241     | DRAFT   |
| Loss gross-up               | Modelled at 3% of delivered energy                         | 0.0876  | 0.146     | ASSUMED |
| Imbalance / backup          | Modelled 2% negative imbalance at THB 4.15/kWh             | 0.0830  | 0.138     | ASSUMED |
| Bond financing carry        | 2% annual bank-guarantee fee; bond principal is refundable | 0.0107  | 0.018     | ASSUMED |

59% generation

23% wheeling

QUANTIFIED PRE-TBD TOTAL

THB 4.78/kWh

+ TBD

≈ THB 7.95bn/year + TBD

GRID-USE BOND — COLLATERAL, NOT TARIFF

THB 889–918m

refundable face value

Recurring carry: 0.011/kWh at 2% bank-guarantee fee; 0.032–0.033/kWh if fully cash-funded at 6%.

KEEP THESE ITEMS AS “TBD”

- Separate data-centre tariff
- Power Development Fund contribution
- Dedicated connection / reinforcement
- Imbalance above 2%, backup capacity and VAT

Every future +0.10 THB/kWh adds ≈ THB 166m/year.

LOAD-BASIS CHECK

If “200 MW” means IT load, PUE 1.30 implies 260 MW at the meter: 2.164 TWh/year, ≈ THB 10.34bn/year pre-TBD and a THB 1.16–1.19bn bond.

# Data Centres – Expert panel - Policy

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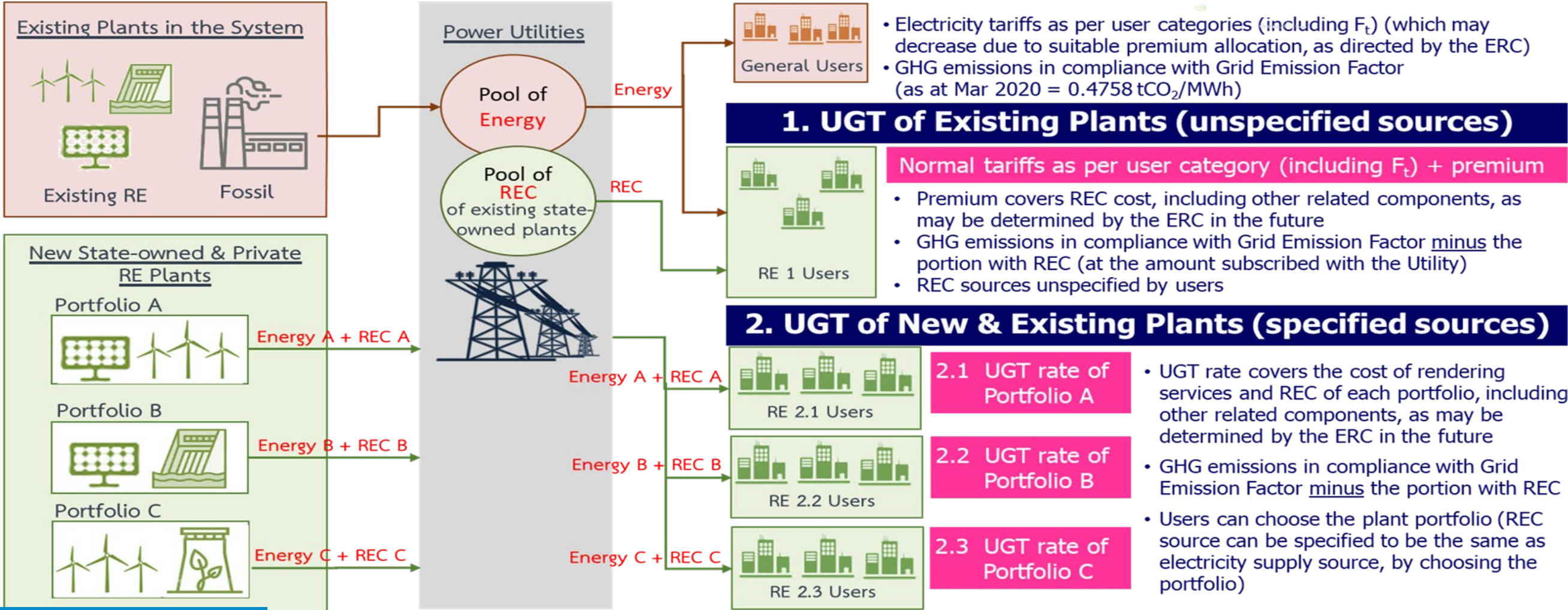
Following two slides –

**Dr Chanchai Amornvipas**

Director of Energy Regulatory Development and Strategic Execution Department.  
Office of Energy Regulatory Commission (ERC)

# Utility Green Tariff: UGT

**\*NEPC resolution (7 Nov 2022) on guidelines for determining Utility Green Tariffs**



## UGT1 (unspecified RE sources)

- launched in July 2025.
- Premium in 2026 for EGAT's Hydro REC = 0.0375 THB/kWh added to the retail tariff.

## UGT2 (specified RE sources)

- Rates approved by ERC on 1st April 2026.
- Rates varied on portfolios and voltage levels.

## Objectives

- To provide an option for power consumers who want to declare the use of electricity generated from RE and to drive the goal of GHG emission reduction of their firms
- To maintain Thailand's comparative advantage in attracting foreign investment to achieve Sustainable Development Goals (SDGs)
- To set Electricity Tariff that reflects the cost of services based on the principle of proportionality and the beneficiary pays principle
- to reduce the cost of subsidizing renewable energy in the power system in the long run

| Key Features — UGT1 vs. UGT2                                                  | UGT1 | UGT2 |
|-------------------------------------------------------------------------------|------|------|
| 1. Use I-REC Tracking Standard                                                | ✓    | ✓    |
| 2. Electricity Price and REC are bundled in one bill                          | ✓    | ✓    |
| 3. Public utility service                                                     | ✓    | ✓    |
| 4. Convenient—no hassle of power purchase agreement (PPA) management          | ✓    | ✓    |
| 5. End Users receive redeemed REC (no price arbitrage)                        | ✓    | ✓    |
| 6. Electricity Supply Agreement contains names of specified RE power plants   | ✗    | ✓    |
| 7. Energy Price varies based on production costs of specified RE power plants | ✗    | ✓    |
| 8. REC and electricity are bundled from the same specified sources            | ✗    | ✓    |
| 9. Help increase new RE power plants in the power system                      | ✗    | ✓    |
| 10. Price Stability (de-coupled from fossil fuel prices; REC price embedded)  | ✗    | ✓    |
| 11. End Users can subscribe partially (not 100% of their electricity demand)  | ✓    | ✗    |



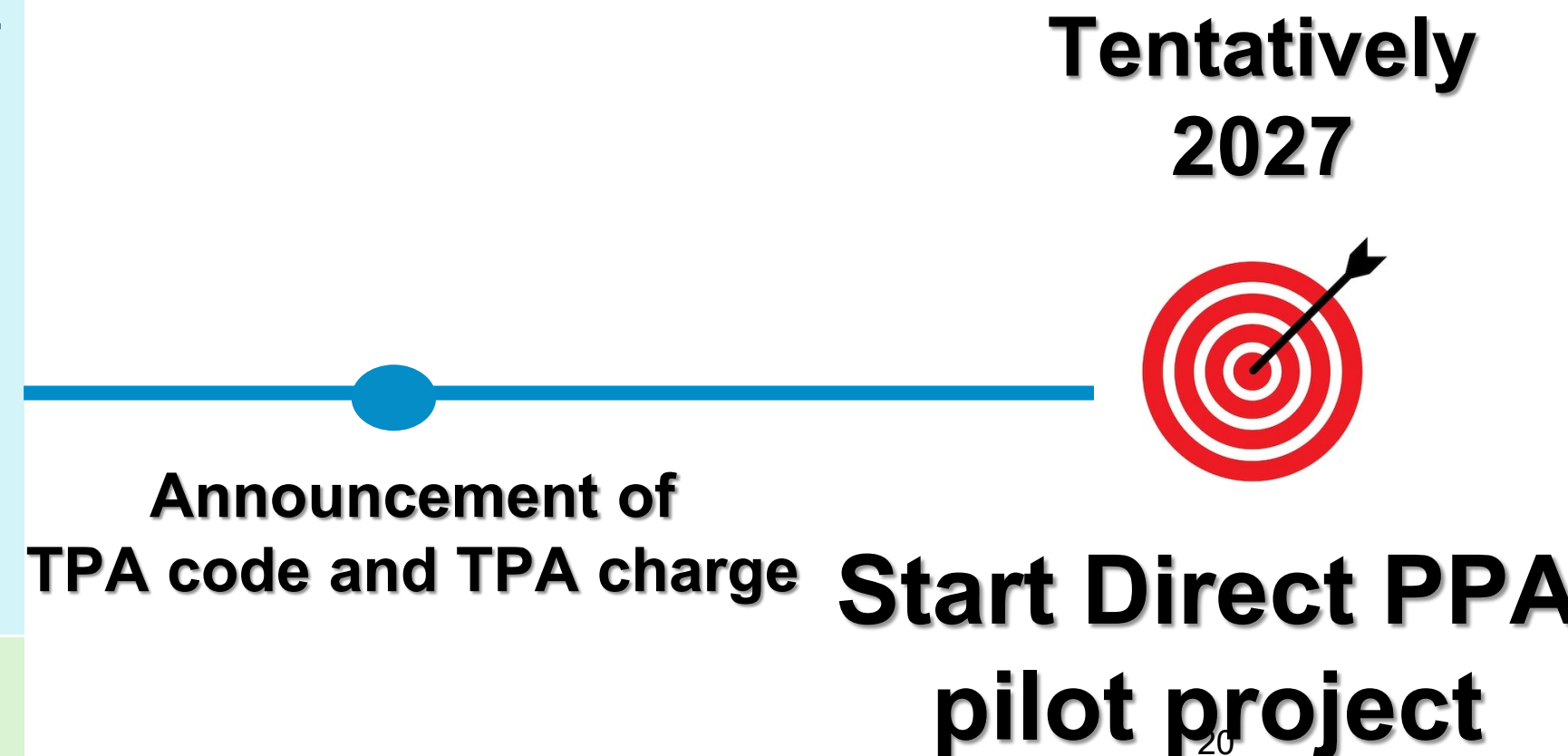
# Third Party Access (TPA) Regime for the Power Sector to provide more choices & promote competition

According to the “**National Energy Reform Plan**” to enhance competitiveness in the power industry and to accommodate the **power sector restructuring** and **deregulation of power trade** in the future

- ERC issued the “TPA Framework Guideline (TPA FG)” in May 2022.
- EGAT, MEA and PEA (as designated licensees) are required to issue TPA Codes in accordance with the TPA FG.

## Direct PPA pilot project

- ✓ **NEPC resolution on 25 June 2024** approving the execution of Electricity Sale Agreements (ESAs) directly between electricity producers and end users, with electricity delivered to end users through the power grid (Third Party Access: TPA)
- ✓ **Max. 2,000 MW** of capacity for trial under the Direct PPA pilot project
- ✓ Target end users ➡ **Data Center** companies using RE supply, as required by their parent companies
- ✓ **NEPC resolution on 15 July 2026** approving Direct PPA scheme for other industries requiring RE supply.



# Data Centres – Expert panel - Policy

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Following three slides –

**Asst Professor Nondh Nuchmorn**; Faculty of Economics Thammasat University

# Thailand Cloud & Data Center Market Overview

Providers & Platforms • Geographic Distribution • Market Growth (2025–2030)

## 1. Cloud Providers & Content Platforms — Local Presence in Thailand

Announcements typically reach Region / Zone level only — underlying facility names are not disclosed

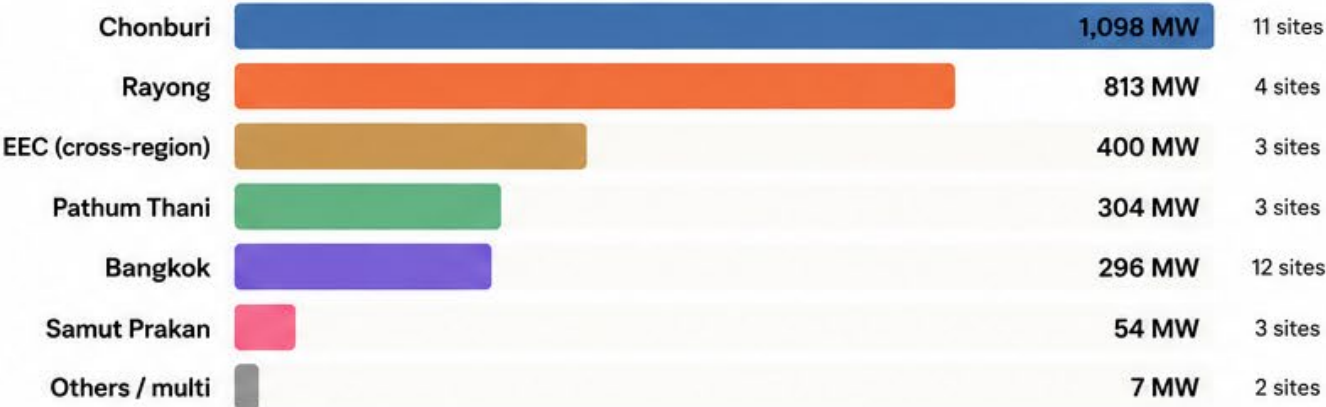
|    |                                                                                                                       |                                                                      |
|----|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| AW | <b>AWS</b><br>ap-southeast-7 (Thailand)<br>Live since Jan 2025 3 AZs                                                  |                                                                      |
| GC | <b>Google Cloud</b><br>asia-southeast3 (Bangkok) + Chonburi DC<br>Live since Jan 2026 3 Zones US\$ 1 B invested       |                                                                      |
| MS | <b>Microsoft Azure</b><br>Thailand cloud region<br>Under construction > US\$ 1 B (2026–28) Hosted at GSA02 + True IDC |                                                                      |
| AI | <b>Alibaba Cloud</b><br>Second Thailand data center<br>Live since Feb 2025                                            |                                                                      |
| TC | <b>Tencent Cloud</b><br>Bangkok · 2 Zones<br>Operational                                                              | <b>HW</b><br><b>Huawei Cloud</b><br>AP-Bangkok region<br>Operational |
| TT | <b>TikTok / ByteDance</b><br>Data hosting project · exact site undisclosed<br>Operational by 2026 THB 126.8 B+        |                                                                      |
| AO | <b>AIS Cloud + Oracle OCI</b><br>Local sovereign cloud platform<br>Operational                                        |                                                                      |

Source: Company announcements, public reports, industry sources (2025)

## 2. Geographic Distribution & Investor Landscape — The two-tier market split and six capital blocs.

### Capacity by province / region

Total MW (includes operational, under construction, approved & planned)

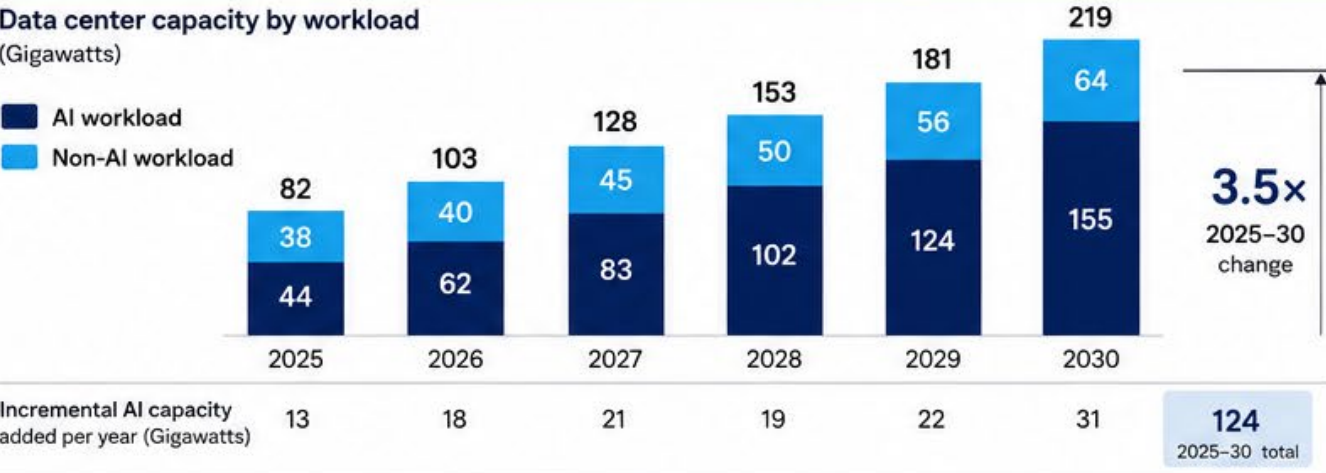


|                                                                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chonburi / Rayong / EEC</b><br>AI & hyperscale battleground<br>DayOne 1 GW platform · Bridge 200 MW · CtrlS 150 MW | <b>Bangkok / Samut Prakan</b><br>Interconnection hub · low latency · finance · cloud region<br>STT · Telehouse · Evolution · GSA01 |
| <b>Pathum Thani</b><br>Navanakorn industrial estate cluster<br>Zenith 200 MW + DAMAC 84 MW                            | <b>Saraburi / Khon Kaen / others</b><br>Secondary expansion nodes<br>INET IDC3 (Saraburi) · IDC4 (Khon Kaen, under construction)   |

## 3. Thailand Data Center Market Growth — AI driving the surge in capacity

### Data center capacity by workload (Gigawatts)

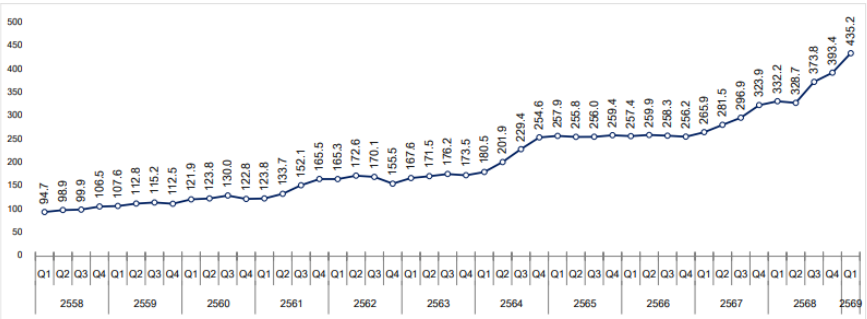
AI workload  
Non-AI workload



Note: Data as of June 2025. Plans and investments are subject to change.

Thailand’s data center boom is accelerating almost unavoidably, making proactive policy action increasingly urgent.

- Land & Resource Challenge
- NDC Target



หมายเหตุ : ปี 2558 เป็นปีฐาน  
ที่มา : ศูนย์ข้อมูลพลังงานไฟฟ้า ภูมิภาคอาเซียน

ตารางที่ 1 ดัชนีราคาที่ดินเปล่าก่อนการพัฒนาในพื้นที่ EEC ไตรมาส 1 ปี 2569

| Index       | Q1/2568 | Q2/2568 | Q3/2568 | Q4/2568 | Q1/2569 |
|-------------|---------|---------|---------|---------|---------|
| พื้นที่ EEC | 332.2   | 328.7   | 373.8   | 393.4   | 435.2   |
| QoQ         | 2.6%    | -1.1%   | 13.7%   | 5.2%    | 10.6%   |
| YoY         | 24.9%   | 16.8%   | 25.9%   | 21.5%   | 31.0%   |

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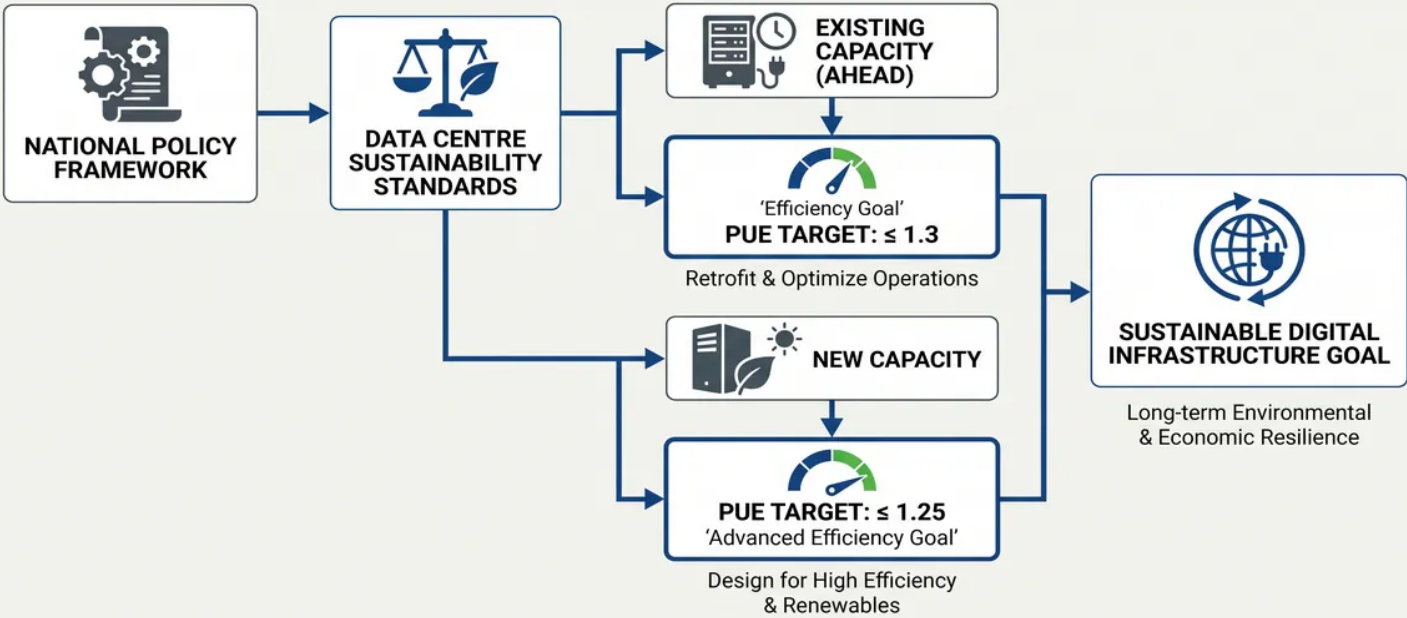
# Driving a Greener Digital Future

## SINGAPORE'S GREEN DATA CENTRE ROADMAP

**Singapore:** Shifted from a 2019–2022 moratorium to selective capacity allocation, with strict green energy and efficiency requirements



### SINGAPORE'S GREEN DATA CENTRE ROADMAP: PUE TARGETS & CAPACITY



Across competing **Southeast Asian** markets, the policy trend is turning from rapid expansion to **more selective, sustainability-linked data center growth**.

Malaysia: (i) Implementation of Data Centre Framework and (ii) Initiative for sustainable data centre development

Data Centre Framework & Green Building Index Data Centre Tool Version 2.0

14 Aug 2025 • 4 minute read

**Malaysia:** Linking investment incentives to PUE and sustainability performance, while prioritizing high efficiency and AI related projects

### Green Building Index Data Centre Tool Version 2.0

### GREENSHIP DATA CENTER: THE PROPOSED RATING SYSTEM OF GREEN DC FOR INDONESIA

#### Greenship DC Rating System

a. The Scope of Greenship DC

The scope of Greenship DC encompasses a variety of Data Center types, from [1] Colocation, [2] Enterprise, [3] Cloud, [4] Edge, and [5] Micro.

Table 1. The Scope of Greenship DC

| GREENSHIP DC |                      |                          |                                         |
|--------------|----------------------|--------------------------|-----------------------------------------|
| No           | Data Center Facility | Sustainability           | Meaning                                 |
| 1            | DCF +                | Greenship NB 2.0         | DC as a new building                    |
| 2            |                      | Greenship EB             | DC as an existing building              |
| 3            |                      | Greenship Neighborhood   | DC consists of several buildings        |
| 4            |                      | Greenship Interior Space | DC in a part of a building              |
| 5            |                      | Greenship Homes          | Small DC in the form of container rooms |

#### Certification Levels

The threshold criteria for Greenship DC certification levels are as follows.

Table 4. Threshold Criteria for Greenship DC Certification Levels

| LEVEL    | PERCENTAGE (%) | RECOGNITION*            |
|----------|----------------|-------------------------|
| PLATINUM | 73 – 100       | Global Leadership       |
| GOLD     | 57 – 72        | National Excellence     |
| SILVER   | 46 – 56        | Outstanding Performance |





## **Key Policy Questions for Thailand**

- Should Thailand establish a **national policy framework** for the green data center industry?
- Which agency should take the lead?
- What should the key policy targets be?
- Which policy instruments should be used: incentives, regulations, taxation, or public investment?
- Which measures should be mandatory, and which should remain voluntary?
- What standards should be adopted?
- Should data center development be subject to zoning?
- Should investment and capacity expansion be phased?
- How can Thailand maximize economic and technological spillovers?

# Question themes – Policy panel

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**THEME ONE: CURRENT POLICIES OR IN PIPELINE**

**THEME TWO: What should the future landscape be?  
What policies are essential?**

## THEME ONE: CURRENT POLICIES OR IN PIPELINE

1. Setting the scene. We have looked at current status of energy policy changes and development, what immediate developments are likely ? what would be added? Would allocation and rationing be inevitable or can that be addressed by economics? \
2. What about water?
3. How if at all are cross subsidies from Data Centres contemplated? (consider the THB 3 / unit for individuals and the higher tariff for DC?)
4. Drawing on previous work supporting ERC on DPPA design and current perspectives (including from ERRRA), what should be the real test of a successful data-centre electricity policy? Beyond simple the availability of sufficient megawatts, or must that power also be reliable, clean/renewable and fairly priced? Should Data Centres be required to source their own renewable energy – perhaps with a forward plan?
5. The July 2026 NEPC policy decisions represent a major step forward. What still separates policy approval from an executable and bankable Direct PPA transaction in Thailand?



## **THEME TWO: What should the future landscape be? What policies are essential?**

6.Can data centres be converted from large inflexible loads into flexible grid resources? What should Thailand learn from international experience on BESS, demand response, locational signals and 24/7 clean-energy procurement?

7.Picking up on an earlier question how should the proposed cost-reflective data-centre tariff, wheeling charges and grid-use bond be designed so that ordinary consumers are protected without double recovery or making Direct PPAs commercially unviable? What can we expect from the Data Centre Policy Committee?

8.Direct PPAs may be short to medium term exceptions to a decentralised, market based but regulated price and availability for energy for many uses, what is a longer term vision for a decentralised industry?



***PREDICTABLE ENERGY*** includes

- availability
- cost
- confidence in transformation to higher renewables demand, and a decentralised sector away from a traditional monopoly